FORMING IONS

10

The noble gases (Group VIII) have atoms with full electron shells; because of this, they are unreactive. All other elements have atoms with incomplete outer shells. By gaining, losing or sharing electrons, atoms can achieve full shells and become stable. Metal atoms tend to lose electrons (which have a negative charge), forming positive ions. Non-metals gain electrons, forming negative ions. If an atom loses one electron it forms an ion with a charge of 1+; if it loses two electrons it forms an ion with a charge of 2+.

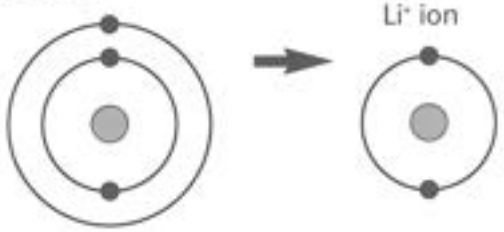
An **ion** is an electrically-charged particle. It has a charge because it contains different numbers of protons (positive) and electrons (negative).

1 Give the charge on an ion formed by an atom that has lost three electrons: _____

2 Complete the diagrams below to show how each atom would form an ion. Include:
- the final electronic configuration (how the electrons are arranged in shells)
- how many electrons have been lost
- the charge on the ion formed.

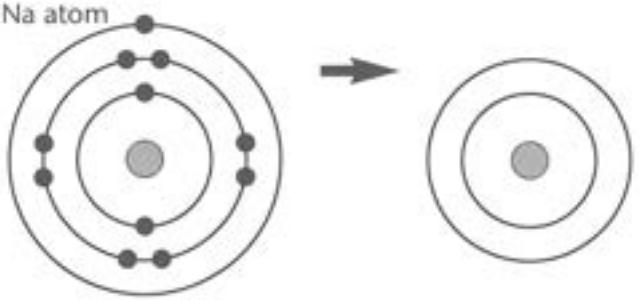
The first one (lithium, Li) has been done for you.

a Li atom

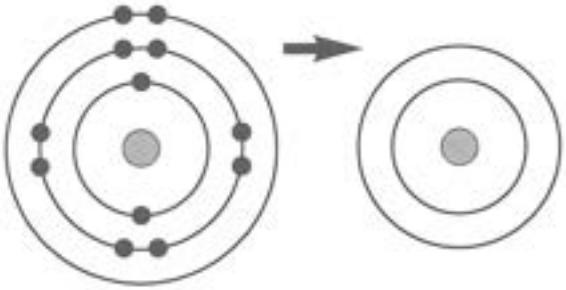


loses one electron

b Na atom



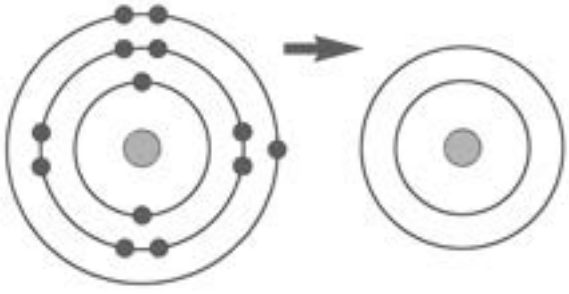
c Mg atom



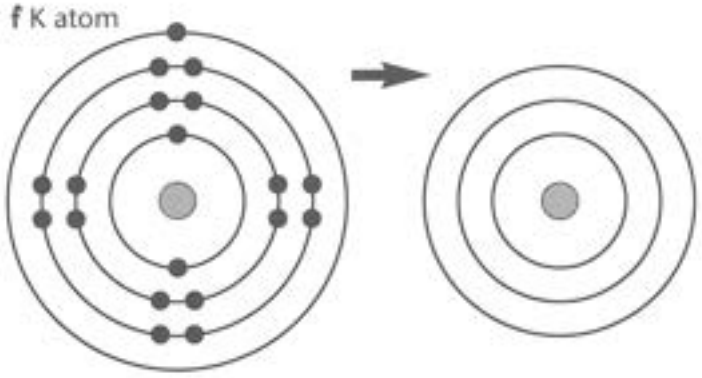
d Be atom



e Al atom



f K atom



3 State two things that the atoms in question 2 have in common.

_____

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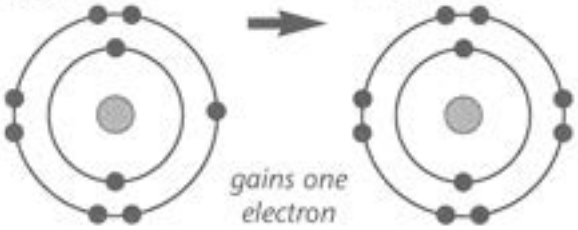
Non-metals, if they form ions, tend to gain electrons. Their ions have more electrons than protons.

4 What is the electrical charge of an ion formed by a non-metal that gains an electron? _____

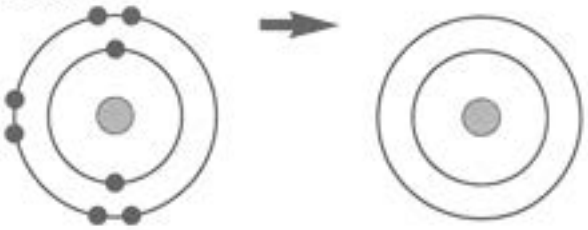
5 Complete the diagrams below to show how each atom would form an ion. Include:
- how many electrons have been gained
- the final electronic configuration
- the charge on the ion.

The first one has been done for you.

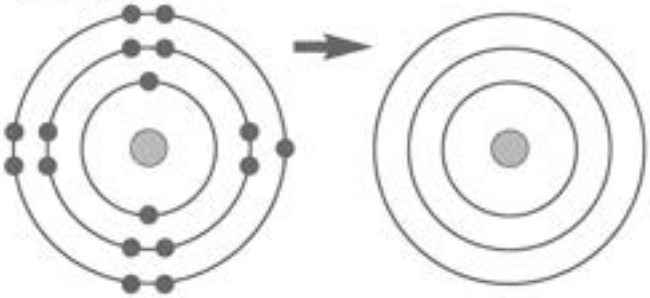
a F atom → F^- ion



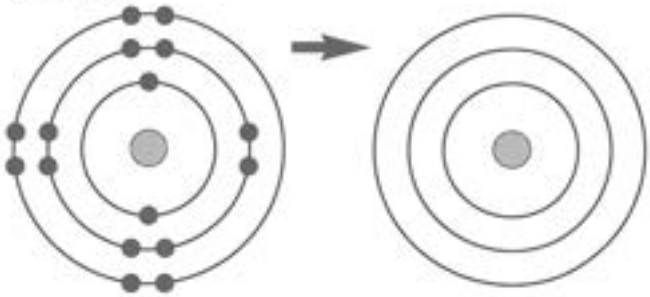
b O atom →



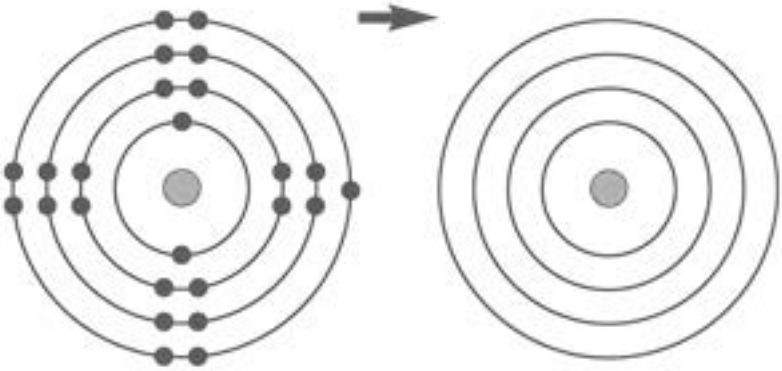
c Cl atom →



d S atom →



e Br atom →



6 State two things that the atoms in question 5 have in common.

_____

7 Write the name of each element and state the group to which it belongs.

a Li	d Be	g F	j S
b Na	e Al	h O	k Br
c Mg	f K	i Cl	

8 What is the relationship between an element's group and the number of electrons it tends to gain or lose?

_____

9 Is it correct to state that a metal often forms a bond with a non-metal? Explain your answer.

_____

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